Introduction to VLSI Design, 2020

Homework 4: pre-layout SPICE deliverable

Requested scope and completed circuit

The original assignment asks for a schematic simulation and then a layout, extraction, and post-layout delay comparison. The requested scope stops at the `.sp` stage. This deliverable therefore contains the complete reusable six-transistor AND netlist, a 1 fF loaded testbench, automated measurement code, and the verified schematic-level data. No layout, DRC/LVS, extracted parasitic circuit, or post-layout result is represented as completed.

The supplied two copies of `homework4.pdf` are preserved in the assignments folder. The HW4 circuit is identical to the HW3 schematic: $F = \overline{AB}$ is the NAND node and $Z = AB$ is the loaded final output. The device sizes, body ties, model, supply, and delay convention are unchanged from the independently verified HW3 design:

Devices	W (μm)	L (μm)
Two NAND PMOS	2.50	0.18
Two NAND NMOS	2.00	0.18
Output PMOS	2.50	0.18
Output NMOS	1.00	0.18

`and2.sp` contains the circuit. The internal node is local to the subcircuit, all NMOS bodies use VSS, all PMOS bodies use VDD, and the external pins are `a b f z vdd vss`. The testbench explicitly places `CL z vss 1f`. Declared source/drain dimensions estimate schematic parasitics and are not an extraction.

Primary testbench

```
.protect
.lib 'D:/cic018.l' TT
.unprotect
.include 'and2.sp'
.temp 25
VDD vdd 0 1.8
VSS vss 0 0
VA a vss PWL(0 0 2n 0 2.02n 1.8 5n 1.8)
VB b vss PWL(0 0 2n 0 2.02n 1.8 5n 1.8)
XAND a b f z vdd vss and2
CL z vss 1f
.tran 1p 5n
.measure tran delay_f TRIG V(a) VAL=0.9 RISE=1 TD=1.9n
+ TARG V(f) VAL=0.9 FALL=1 TD=1.9n
.measure tran delay_z TRIG V(a) VAL=0.9 RISE=1 TD=1.9n
+ TARG V(z) VAL=0.9 RISE=1 TD=1.9n
.end
```

The distributed `testbench.sp` also records output precision and before/after voltages. The model path is a local prerequisite.

Verified pre-layout worst-case results

These are the same actual HSPICE measurements as HW3, because HW4 reuses the exact same circuit. They are not a second or post-layout simulation. CIC018 TT, 1.8 V, 25 C, 1 fF, 20 ps input ramp, and the 50-percent crossing convention apply.

Node	Direction	Input transition	Delay (ps)
F	HL	`00->11`	29.4932
F	LH	`11->10`	41.8569
Z	HL	`11->10`	63.7000
Z	LH	`00->11`	47.6815

All six output-changing transitions were covered in both directions. Before/after levels pass the 1.6 V high and 0.2 V low bands. Tightening the timestep to 0.2 ps provides the accompanying convergence evidence. Raw license details and the process model are excluded from the repository.

Comparison and corrections

Two supplied HW4 submissions were reviewed: J.-M. Wu's PDF and `pre-sim.sp`, and S.-Y. Sun's report and `1.inv.sp`. Wu's HW4 pre-simulation fixes the upper NMOS body to VSS, but still uses 3.3 V on `N_18/P_18` and omits the 1 fF load in that standalone file. Sun's listed schematic follows the same six-transistor topology. Their layout and extracted-netlist artifacts are background only because the requested scope stops before layout. Reported faster post-layout values are not imported into this pre-layout design.

Reproduction and consistency check

Run `python simulate.py -model D:/cic018.l -raw-dir <scratch>`. The copied script performs the full six-transition sweep and exports the same schema as HW3. The included `provenance.json` records that the checked-in result CSVs are reused from HW3 and gives the SHA-256 hash of the identical netlist. Regenerating HW4 with the same model and simulator should reproduce them within simulator precision. For a single primary run, invoke HSPICE directly on `testbench.sp` from this directory.